

SUBNETTING BASICS

1. Given the network 192.168.10.0/25, calculate the subnet mask, network address, broadcast address, and the range of usable IP addresses.

-> For **192.168.10.0/25**:

Step 1: Subnet mask

/25 means 25 network bits and 7 host bits.

Subnet mask: 255.255.255.128

Step 2: Network address

The given network starts at:

192.168.10.0

Step 3: Broadcast address

A /25 subnet contains:

$2^7 = 128$ total addresses.

So the range is .0 through .127.

Broadcast address: 192.168.10.127

Step 4: Usable IP range

Exclude the network address (.0) and broadcast address (.127).

Usable IP addresses: 192.168.10.1 – 192.168.10.126

Final Answer

Item	Value
Network	192.168.10.0/25
Subnet Mask	255.255.255.128
Network Address	192.168.10.0
Broadcast Address	192.168.10.127
Usable IP Range	192.168.10.1 – 192.168.10.126
Total Addresses	128

Item	Value
Usable Addresses	126

2. Your company wants to split the 10.0.0.0/24 network into four equal subnets for different departments (HR, Sales, Tech, Admin). List the network address, broadcast address, and usable IP range for each subnet.

-> To divide **10.0.0.0/24** into **4 equal subnets**, borrow 2 host bits:

- Original: /24
- New prefix: /26
- Subnet mask: **255.255.255.192**
- Each subnet has **64 total addresses**
- Each subnet has **62 usable addresses**

Subnet Allocation

Department	Network Address	Usable IP Range	Broadcast Address
HR	10.0.0.0/26	10.0.0.1 – 10.0.0.62	10.0.0.63
Sales	10.0.0.64/26	10.0.0.65 – 10.0.0.126	10.0.0.127
Tech	10.0.0.128/26	10.0.0.129 – 10.0.0.190	10.0.0.191
Admin	10.0.0.192/26	10.0.0.193 – 10.0.0.254	10.0.0.255

Final Summary

10.0.0.0/24

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|— HR → 10.0.0.0/26

|— Sales → 10.0.0.64/26

|— Tech → 10.0.0.128/26

└ Admin → 10.0.0.192/26

Each department therefore gets **62 usable IP addresses**.

3. A startup needs 60 usable IP addresses for its team. What is the smallest CIDR subnet you can assign, and what are the network address, broadcast address, and usable IP range for this subnet?

-> To support **60 usable IPv4 addresses**, we need to find the smallest subnet that provides at least 60 usable addresses.

Step 1: Find the required host bits

Formula:

$$\text{Usable addresses} = 2^h - 2$$

Try 6 host bits:

- $2^6 - 2 = 64 - 2 = 62$ usable addresses 

So we need **6 host bits**.

$$\text{CIDR prefix} = 32 - 6 = /26$$

Therefore, the smallest subnet is **/26**.

Step 2: Choose a subnet

Using **192.168.1.0/26** as an example:

Item	Value
CIDR	192.168.1.0/26
Subnet Mask	255.255.255.192
Total Addresses	64
Usable Addresses	62
Network Address	192.168.1.0
Usable IP Range	192.168.1.1 – 192.168.1.62
Broadcast Address	192.168.1.63

Answer

Smallest CIDR: /26

Network: 192.168.1.0

Broadcast: 192.168.1.63

Usable range: 192.168.1.1 – 192.168.1.62

This gives **62 usable addresses**, which is enough for the startup's 60 devices while being smaller than a /25 subnet.

4. For the public WiFi at a Zomato restaurant, you are given the IP block 172.16.5.0/26. Calculate the subnet mask, network address, broadcast address, and the first and last usable IPs.

Hint: /26 gives you 64 addresses per subnet.

-> For the IP block **172.16.5.0/26**:

- **CIDR:** /26
- **Subnet mask:** 255.255.255.192
- **Total addresses:** 64
- **Network address:** 172.16.5.0
- **Broadcast address:** 172.16.5.63
- **First usable IP:** 172.16.5.1
- **Last usable IP:** 172.16.5.62
- **Usable IP addresses:** 62

How it is calculated

A /26 leaves **6 host bits**:

$$2^6 = 64 \text{ addresses}$$

The range is therefore:

172.16.5.0 → 172.16.5.63

The first address is the **network address**, and the last is the **broadcast address**, leaving .1 through .62 for devices.

Parameter	Value
Network	172.16.5.0/26
Subnet Mask	255.255.255.192
Network Address	172.16.5.0
First Usable	172.16.5.1

Parameter	Value
Last Usable	172.16.5.62
Broadcast	172.16.5.63

5. You receive the following scenario: Flipkart's warehouse needs three subnets supporting at least 30 devices each. Using CIDR notation, assign suitable subnets from the 192.168.1.0/24 range, and specify the network address, broadcast address, and usable IP range for each subnet.

-> To support **at least 30 devices per subnet**, we need a subnet with **30 usable IP addresses**.

Step 1: Choose the subnet size

A /27 subnet provides:

- Total addresses: $2^{32-27} = 32$
- Usable addresses: **32 - 2 = 30**
- Subnet mask: **255.255.255.224**

From **192.168.1.0/24**, we can divide it into /27 subnets.

Step 2: Assign three subnets

Subnet	CIDR	Network Address	Usable IP Range	Broadcast Address
Subnet 1	192.168.1.0/27	192.168.1.0	192.168.1.1 – 192.168.1.30	192.168.1.31
Subnet 2	192.168.1.32/27	192.168.1.32	192.168.1.33 – 192.168.1.62	192.168.1.63
Subnet 3	192.168.1.64/27	192.168.1.64	192.168.1.65 – 192.168.1.94	192.168.1.95

Final Answer

The three suitable subnets are:

1. **192.168.1.0/27** → usable: 192.168.1.1–30

2. **192.168.1.32/27** → usable: 192.168.1.33–62

3. **192.168.1.64/27** → usable: 192.168.1.65–94

Each subnet supports **exactly 30 usable devices**, satisfying the requirement.